Freight Load Pricing Prediction

Technical Assessment Report

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1. Executive Summary

This report documents the approach used to predict posted freight rates (posted_rate) from load-level features. The development data (train-test.csv, 48,000 loads, January-October 2025) was explored, cleaned, and used to engineer a feature set covering distance, seasonality, market-condition signals, equipment, and origin/destination city. Because the two required prediction targets - validation.csv (November 2025) and the December 2025 chart inputs - both fall after the end of the training window, model validation was built around a time-based split rather than a random one, so that reported accuracy reflects genuine forward-in-time forecasting.

A log-target Linear Regression model, trained on engineered features with a small number of flagged noisy rows excluded, gave the best holdout accuracy of every model tested (MAE $120.34, MAPE 5.57%, R^2 0.830) while remaining fully interpretable. A second model with the same architecture, trained without the market_index and quote_signal columns, was built specifically for the December chart, which never provides those two columns; this fallback model is measurably less accurate (MAPE 7.29%) - a real, quantified cost of the missing data rather than a modeling shortfall.

2. Data Overview

Three files were provided, spanning three consecutive, non-overlapping time windows:

File	Rows	Date range	Notes
train-test.csv	48,000	Jan 1 – Oct 31, 2025	Labeled development data (posted_rate present)
validation.csv	12,000	November 2025	Unlabeled; final predictions required
december-chart-inputs.csv	31	December 2025	Fixed lane/equipment/weight; no market_index or quote_signal

Table 1. Source files and their time coverage.

Because both prediction targets sit strictly after the training window, the central question this report addresses is how the model is validated to make sure it generalizes forward in time - covered next.

3. Train / Validation Split Strategy

A naive random split of train-test.csv would let the model “peek” at rows from the same weeks it is later tested on, and would overstate accuracy relative to the real task: predicting rates for a month the model has never seen. Instead, a time-based split was used, holding out the last 6 weeks of the training window (September 19 - October 31, 2025) as an internal holdout set, with the remaining data used to fit candidate models.

Split	Rows	Date range	Share
Dev (fit)	41,468	Jan 1 – Sep 19, 2025	86.4%
Holdout (evaluate)	6,532	Sep 19 – Oct 31, 2025	13.6%

Table 2. Time-based dev/holdout split used for model selection.

This mimics the real forecasting gap the model faces: holdout data is genuinely in the future relative to the data the model was fit on, the same way November and December are relative to the full training set. All model comparisons and the outlier-handling decision in this report are based on holdout performance, never on the dev set the models were fit on.

Categorical encoders (smoothed target encoding for pickup/delivery cities) and imputation statistics (median weight and market_index) were fit only on the dev portion when producing the holdout evaluation, to avoid leaking holdout information into the encoding - city-level target encoding in particular can leak the target if not handled carefully. Out-of-fold encoding (5-fold) was used for the dev set's own encoded values for the same reason.

Once a final model was selected using this split, both the full model and the December fallback model were refit on all 48,000 labeled rows (using encoders refit on the full training set) before generating the actual validation and December predictions - the holdout split exists to select and validate the approach, not to reduce the data used for the final submission.

4. Data Quality Issues Identified & Addressed

- **Negative weight values:** 292 training rows (0.6%) had negative weight values with a range that mirrors the positive range almost exactly, consistent with a sign-flip data-entry error rather than genuinely negative weight. Corrected with abs(weight).
- **Rate-per-mile outliers:** 671 training rows (1.4%) had a rate-per-mile far outside the tight band the rest of the data follows, independent of the weight issue (only 9 rows overlapped). These were flagged and excluded from model training after confirming on the holdout set that exclusion improved accuracy.
- **Missing values:** weight (0.6% train / 1.4% validation) and market_index (0.8% train / 2.1% validation) had missing values, flat across equipment type and month - consistent with missing-completely-at-random. A missing-indicator flag was added for each before median imputation.
- **Unseen cities in validation:** 8 cities appear in validation.csv that never appear in train-test.csv (Knoxville, Jackson, Charlotte, Norfolk, Allentown, Chicago, Laredo, San Diego). The city encoder falls back to the global mean rate for any unseen city, verified directly against these 8 cities rather than assumed.
- **Missing market signal columns in December file:** december-chart-inputs.csv never includes market_index or quote_signal, the two columns shown to carry most of the non-distance predictive signal. Addressed with a dedicated fallback model trained without those columns (Section 7).

5. Feature Engineering Summary

- **Distance:** Raw distance plus log1p(distance), since rate-per-mile declines non-linearly as distance increases.
- **Temporal:** Cyclical sin/cos encodings of day-of-year and day-of-week, which extrapolate sensibly into November/December, unlike raw month numbers or one-hot months the model has never seen; plus a linear days_since_start trend term.
- **Market signals:** market_index × distance and quote_signal × distance interaction terms, since both signals correlate weakly with rate on their own (~0.03) but strongly once multiplied by distance (0.88–0.90) - they scale the base rate rather than shifting it.
- **Equipment:** One-hot encoding - low cardinality (3 classes) and identical across all files.
- **Pickup / Delivery:** Smoothed target encoding (shrunk toward the global mean by sample size) for pickup and delivery cities, fit out-of-fold on training data to avoid leakage, with an explicit global-mean fallback for unseen cities.
- **Geographic coordinates:** pickup_lat/lon and delivery_lat/lon were dropped - confirmed fixed per city (zero within-city variance) and therefore redundant with the city name and with distance (0.9995 correlation with great-circle distance computed from them).

6. Model Selection

Four model families - Linear Regression, Ridge, Random Forest, and HistGradientBoosting - were each trained on both the raw target and a log1p(posted_rate) target, and evaluated on the time-based holdout set described in Section 3.

Model	Target	MAE	RMSE	MAPE	R^2
Linear Regression	log	$124.89	$630.83	5.76%	0.829
Ridge	log	$125.26	$630.97	5.77%	0.829
Random Forest	log	$158.83	$645.51	7.22%	0.821
HistGradientBoosting	log	$129.49	$632.91	5.93%	0.828
Linear Regression	raw	$147.12	$633.81	8.05%	0.827
HistGradientBoosting	raw	$147.78	$636.50	6.70%	0.826

Table 3. Holdout performance by model family and target transform (best 6 of 8 combinations shown).

The log target consistently outperformed the raw target across every model family, consistent with the right-skewed distribution of posted_rate found during EDA. Model families performed similarly once the target was log-transformed - the engineered features (log-distance, market-signal interactions, city encodings) appear to capture most of the usable non-linear structure, leaving comparatively little advantage for the more flexible tree-based models.

Excluding the 671 flagged rate-per-mile outlier rows from training (Section 4) was then tested on the log-target models:

Model	Outliers	MAE	RMSE	MAPE	R^2
Linear Regression	kept	$124.89	$630.83	5.76%	0.829
Linear Regression	removed	$120.34	$628.58	5.57%	0.830
Random Forest	kept	$158.83	$645.51	7.22%	0.821
Random Forest	removed	$124.93	$633.17	5.56%	0.828
HistGradientBoosting	kept	$129.49	$632.91	5.93%	0.828
HistGradientBoosting	removed	$137.54	$631.10	6.28%	0.829

Table 4. Effect of excluding flagged outlier rows from training (holdout set is unchanged in every row).

Excluding the outlier rows improved Linear Regression and Random Forest, and slightly hurt HistGradientBoosting - consistent with the flagged rows being noise rather than a pattern worth fitting for most model types.

7. Final Model & Interpretation

Linear Regression on log1p(posted_rate), trained with the flagged outlier rows excluded, was selected as the final full-feature model. It tied for the best or was the best result on every metric in Tables 3–4, and unlike the tree-based alternatives, every prediction can be explained directly as a sum of coefficients × features.

Because features are on very different scales (e.g. distance spans 0–3,500 while month_sin spans −1 to 1), raw regression coefficients are not directly comparable. The table below instead reports each feature's approximate effect on log(rate) across its full observed range in the training data (|coefficient| × range) - a fairer view of relative influence.

Feature	Approx. effect on log(rate)
log_distance	3.30
weight	0.13
market_index	0.12
distance	0.11
days_since_start	0.08
equip_Dry_Van	0.07
quote_signal	0.07
equip_Reefer	0.05
quote_signal × distance	0.04

Table 5. Scale-adjusted feature effects, largest first (full table of 19 features is in the modeling notebook, Section 8).

log_distance dominates by a wide margin - consistent with the EDA finding that distance alone correlates 0.91 with posted_rate - followed by weight, market_index, and the linear time trend. This matches domain intuition: distance sets the base price, and market conditions and equipment scale it from there.

8. No-Market Fallback Model (December)

december-chart-inputs.csv never includes market_index or quote_signal, so the full model above cannot be applied to it. A second model - identical architecture (Linear Regression, log target, outliers excluded) - was trained on the same rows with those two columns and their interaction terms removed, and evaluated on a matching reduced-column copy of the holdout set, so the comparison below isolates the effect of the missing columns rather than a difference in model type.

Model	MAE	RMSE	MAPE	R^2
Full model (with market signals)	$120.34	$628.58	5.57%	0.830
No-market fallback model	$167.49	$645.07	7.29%	0.821

Table 6. Holdout accuracy cost of predicting without market_index and quote_signal.

The fallback model is measurably, but not drastically, less accurate - MAPE rises from 5.57% to 7.29%. This is reported as a known limitation of the December predictions rather than a modeling gap: no model can recover information that isn't present in its inputs.

9. Validation Predictions

The full model, refit on all 47,329 rows remaining after excluding flagged outliers, was applied to all 12,000 rows of validation.csv. Predictions were sanity-checked before saving: exactly 12,000 rows, no missing or non-positive values, load_id coverage matching the template exactly.

Metric	Value
Rows predicted	12,000
Minimum predicted_rate	$204.33
Maximum predicted_rate	$6,900.47

Metric	Value
Mean predicted_rate	$2,418.75

Table 7. Summary of validation_predictions.csv.

10. December 2025 Prediction Chart

The no-market fallback model was used to predict predicted_rate for all 31 days of December 2025 on the fixed lane specified in december-chart-inputs.csv (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb - only the date changes across rows). The completed file was validated and charted using the provided score.py, reproduced below exactly as generated.

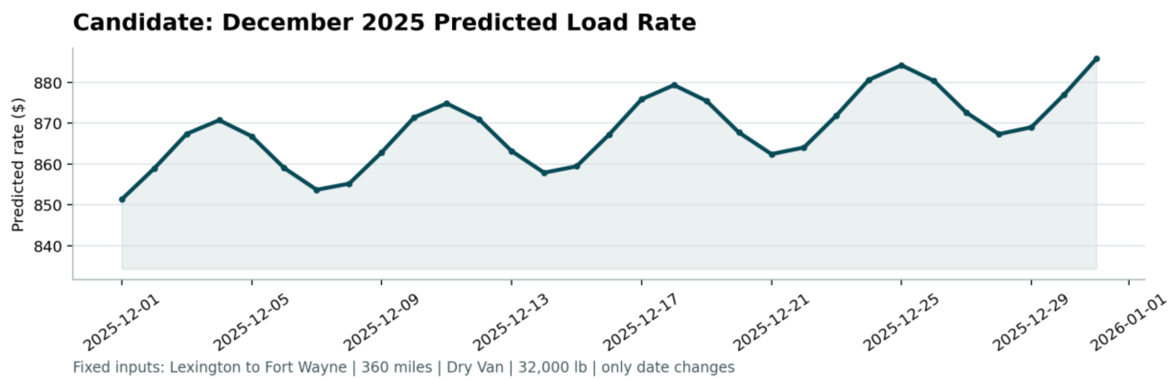

Figure 1. candidate_december.png, produced by score.py from the submitted December predictions.

Predicted rates range from $851.35 to $885.78 across the month. The curve shows a mild weekly cycle (from the day-of-week features) layered on a slight upward drift toward year-end, consistent with the seasonal uptrend the EDA observed heading into the winter months on the years' worth of training data available.

11. Limitations & Future Work

- **No market signal for December:** December predictions rely on a model trained without market_index/quote_signal, which are the two strongest signals after distance. A useful follow-up would be estimating those two columns for December from the seasonal pattern observed in training, and comparing predictions from that approach against the current fallback model.
- **Extrapolation risk:** Both November (validation) and December fall outside the observed Jan-Oct training window, so the model is extrapolating the seasonal cycle rather than interpolating within it. The time-based holdout (Sep–Oct) gives a reasonable proxy for this, but true out-of-sample accuracy for November/December can only be confirmed once ground truth is available.
- **Model tuning:** Linear Regression was chosen for its holdout accuracy and interpretability. Tree-based models performed similarly on holdout and might respond differently to a wider range of hyperparameters or additional feature engineering; this was not exhaustively tuned given the scope of this assessment.
- **Outlier threshold:** The rate-per-mile outlier threshold ([1.0, 3.5] $/mi) was chosen from the training distribution rather than a rigorous statistical procedure. A more principled approach (e.g. an isolation-forest or a per-lane threshold) could refine this.